# Madan Bhandari Memorial College

New Baneshwor, Kathmandu

## Mid-Term Examination 2080

## 2080/11/23; 7:00 AM

Subject : MTH 117 Mathematics I
BScCSIT First Semester
Institute of Science and Technology

Full Marks : 60
Pass Marks : 24
Time : 3 hrs.

***Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.***

### Group A

**Attempt any TWO Questions** **(2×10=20)**

1) Let $f(x) = 2x + 1$, $g(x) = \sin x$; where $f,g: R \to R$. Find $(f+g)(x)$, $(f-g)(x)$, $(fg)(x)$ and $(\frac{f}{g})(x)$. Also find their domain. [2+2+2+2+2]

2) a) Find the equation of the tangent line to the curve $y=4x-3x^2$ at the point $(2,-4)$. [5]
   b) State Roll's Theorem. Show that the function f(x) satisfies Roll's Theorem. [5]
   $f(x) = x^3 - x^2 - 6x + 2$, $[0, 3]$

3) a) Define initial value problem. Solve that initial value problem of $y'+2y=3$, $y(0)=1$ [6]
   b) Evaluate the indefinite integral: $\int x^3 \cos(x^4 + 2)\, dx$ [4]

### Group B

**Attempt any EIGHT Questions** **(8×5=40)**

4) Find the domain and range of $f(x) = -x^2 + 4x - 3$. [2+3]

5) Define limit at a point. Prove that $\lim_{x \to 0} \frac{|x|}{x}$ does not exist. [1+4]

6) Find all the asymptotes of the graph $f(x) = \frac{2x^2+5x+7}{x-1}$.

7) Calculate the indeterminate form: $\lim_{x \to 0^+} (1 + \sin 4x)^{\cot x}$

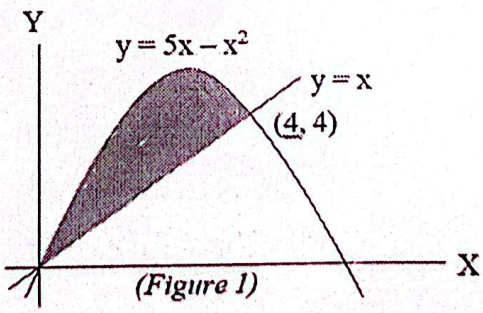
8) Sketch the curve $f(x) = x^3 + x$.

9) Determine the type of improper integral given below and evaluate: $\int_0^1 \ln x \, dx$

10) Find the area of the shaded region in figure 1.

11) The region R enclosed by the curves $y = x$ and $y = x^2$ is rotated about the **x – axis**. Find the volume of the resulting solid.

12) Find the solution of $y'' + 6y' + 9y = 0$, $y(0) = 2$, $y'(0) = 1$.



*(Figure 1)*

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# Madan Bhandari Memorial College

New Baneshwor, Kathmandu

## Mid-Term Examination 2080

## 2080/11/20; 7:00 AM

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|---|---|---|---|
| Subject : CSC 115 C Programming | | Full Marks | : 60 |
| BScCSIT First Semester | | Pass Marks | : 24 |
| Institute of Science and Technology | | Time | : 3 hrs. |

***Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.***

### Group A

**Attempt any TWO Questions:** **(2x10=20)**

1) Write a program to find transpose of a matrix using concept of function.

2) Why is switch case statement required? Explain with example. Identify and list the errors in the following code.

```
#include<stdio.h>
#include<conio.h>
void sum(int,int,int)  ; missing
int main()
{
int a,b,c   ; is missing
print("Enter three numbers);  " missing
scanf("%d%d%d, a, &b, &c);
printf("Sum of three numbers is %c", sum(a, b, c));
return -1;   — error in line 10
}
void sum (int x, int y, int z);   ; is mis
{
int sum;
sum = a + b + c;
return sum;
};
```

9 Syntax r.d

3) What is preprocess directive? Create a structure named department having department id, department name, no. of employees as members. Input information of 5 department and display the details of department having no of employees between 30 to 40.

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## Group B

**Answer any EIGHT questions:** (8x5=40)

4) List different types of operators and explain any three of them.
5) Draw a flowchart to check whether a number is prime or not.
6) Write a program to find maximum element of 2D array.
7) Write a program to find factorial of given number entered by user.
8) Difference between pass by value and pass by reference with example.
9) Write a program to find Fibonacci series using recursion.
10) Explain use of printf and scanf in C programming.
11) Write short notes on:
    a) Compiler and Interpreter
    b) Continue and break statement
12) Differentiate betn. do-while & while loop.

# Madan Bhandari Memorial College

New Baneshwor, Kathmandu

## Mid-Term Examination 2080

**2080/11/21; 7:00 AM**

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Subject : CSC 114 Introduction to Information Technology
BScCSIT First Semester
Institute of Science and Technology

Full Marks : 60
Pass Marks : 24
Time : 3 hrs.

***Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.***

## Group A

**Attempt any TWO Questions.** (2x10=20)

1) Explain hierarchy of different memory types along with figure. What different types are of read only memory?
2) Compare system software with application software. Explain different types of system software in brief.
3) List some benefits of using computer networks. Explain different types of data transmission media in detail.

## Group B

**Attempt any EIGHT Questions.** (8x5=40)

4) What is digital computer? Compare digital computers with analog computers.
5) Explain each component of CPU in brief.
6) Define hard copy and soft copy devices. List any six human data entry devices.
7) What is ASCII? Convert 0.2345 from base 10 to base 2.
8) Explain wearable computing in brief.
9) Define database. What are the benefits of using databases?
10) What are different characteristics of multimedia?
11) Define cryptography. Why do we use firewall?
12) Write short notes on:
    a) BigData
    b) Instruction Cycle

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# Madan Bhandari Memorial College

New Baneshwor, Kathmandu

## Mid-Term Examination 2080

## 2080/11/24; 7:00 AM

Subject : PHY 118 Physics
BScCSIT First Semester
Institute of Science and Technology

Full Marks : 60
Pass Marks : 24
Time : 3 hrs.

***Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.***

### Group A

**Long Answer Questions (Attempt any TWO)** **(2x10=20)**

1) Explain the theory of black body radiation. Why this theory needs quantum mechanical interpretation? How this interpretation experimentally successful? Explain.
2) Setup differential equation for an oscillation of a spring using Hooke's law and Newton's second law. Find the general solution of this equation and hence the expressions for period, velocity and acceleration of oscillation.
3) Describe Frank Hertz experiment. Discuss its result and outline limitations.

### Group B

**Short Answer Questions (Attempt any EIGHT)** **(8×5=40)**

4) Discuss magnetic dipole moment. What is its effect on atom and on molecules? Explain.
5) Explain Bloch theorem? Discuss its use in Kronig-Penny model and hence in band theory.
6) An oscillating block of mass 250 g takes 0.15 sec to move between the endpoints of the motion, which are 40 cm apart. Find (a) frequency, (b) amplitude of the motion, and (c) force constant of the spring.
7) A current of 50A is established in a slab of copper 0.5cm thick and 2 cm wide. The slab is placed in a magnetic field B of 1.5 T. The magnetic field is perpendicular to the plane of the slab and to the current. The free electron concentration in copper is $8.4\times10^{28}$ electrons/$m^3$? What will be the magnitude of the gall voltage across the width of the slab?)
8) What are (a) the energy, (b) the momentum, and (c) the wavelength of the photon that is emitted when a hydrogen atom undergoes a transition from the state n = 4 to n = 2?

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